A-MEM Reproduction on a LoCoMo Subset with a Memory Evolution Ablation

FTEC5660 Reproducibility Work

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Repository https://github.com/LeoLIU1233/A-mem

Result The full pipeline runs successfully and gives a positive overall signal for Memory Evolution, but the result is only a partial behavioral reproduction. DeepSeek, a 25-question subset, and k=10 prevent direct comparison with the paper's GPT-4o-mini table.

1 Reproduction target

The target is the paper's comparison between full A-MEM and A-MEM with Memory Evolution removed. The reported GPT-4o-mini Multi-Hop F1 is 27.02 for the full model and 21.35 without ME. Temporal F1 is 45.85 and 31.24, respectively. This comparison isolates one architectural component and has a directly measurable effect.

I used the authors' evaluation repository rather than rebuilding the memory system from the paper description. The baseline is WujiangXu/AgenticMemory at commit f9a173dec325d84fcc97e7828775c7b241c33466. The reference system implementation is agiresearch/A-mem.

2 Reproduction setup

The evaluation retains note construction, embedding retrieval, link generation, and Memory Evolution. It uses the repository's robust plain-text path so the pipeline can run with an OpenAI-compatible DeepSeek endpoint.

Condition	Completed run	Paper target
Model	deepseek-chat	GPT-4o-mini
Conversations	1 of 10	LoCoMo evaluation set
Dialogue context	First 120 turns	Full conversations
Questions	5 per category; 25 total	Full benchmark
Retrieval k	10 for every category	40/40/50/50/40
Seeds	42, 43, 44	Not reported
Adversarial scoring	Current Not-mentioned reference	Legacy reference

The released locomo10.json file contains 10 conversations, 1,986 QA pairs, and 5,882 turns. Its SHA-256 is 553CD5A15E25F2CECCC6ED185221EBA645080C93E5B91087560A91AA5961F365. The paper reports 7,512 QA pairs, so the available data count is a material reproduction condition.

3 Isolated modification

The main scientific change is the link_only mode. Full mode builds a note, retrieves neighbors, asks the LLM whether to establish links, and updates neighbor context and tags when required. Link_only keeps note construction, retrieval, the evolution decision, and link creation, but skips neighbor context and tag updates.

DeepSeek support, deterministic category sampling, category-specific k, cache isolation, and tests are control and compatibility changes. The scientific treatment is the removal of Memory Evolution.

4 Results

Mode	Seed 42 F1	Seed 43 F1	Seed 44 F1	Mean F1	SD
Full A-MEM	39.68	43.52	42.10	41.76	1.94
Link only, w/o ME	40.42	42.01	32.32	38.25	5.20

Mean BLEU-1 was 37.83 for full A-MEM and 35.32 for link-only A-MEM. The paired F1 differences by seed were -0.75, +1.51, and +9.78 points. The mean paired gain was +3.51 points with a standard deviation of 5.54.

Category	Full mean F1	Link-only mean F1	Paired gain
Multi-Hop	44.24	28.69	+15.56
Temporal	11.59	12.62	-1.03
Open-Domain	21.15	13.35	+7.81
Single-Hop	45.17	56.60	-11.43
Adversarial	86.67	80.00	+6.67

5 Interpretation

The Multi-Hop gain is consistent with the intended mechanism. Links and updated neighbor descriptions can expose evidence from different dialogue sessions. The Temporal result does not reproduce the paper's large advantage, and Single-Hop is better without Memory Evolution in this subset.

The variation is substantial. Multi-Hop has a paired standard deviation of 16.78 points, and one seed makes the full model slightly worse overall. With five questions per category, one changed answer moves a category score by 20 points. The positive mean is therefore evidence for a possible effect, not a decisive estimate.

6 Reproducibility boundaries

- DeepSeek replaces GPT-4o-mini, so absolute scores are not directly comparable with the paper.
- The completed run uses k=10 instead of the paper's category-specific retrieval values.
- The remote model does not receive the local seed, so trials remain stochastic.
- The corrected Adversarial reference differs from the paper's legacy reference.
- The sample contains one conversation and 25 questions.

7 Debugging record

The system implementation repository directs benchmark reproduction to WujiangXu/AgenticMemory, so evaluation switched to that repository. The original path expects structured JSON output; the robust plain-text path was used for DeepSeek. Category-specific retrieval settings were not available through the original command line, so retrieve_k_map was added.

A local fake OpenAI-compatible server tested note construction, query generation, answer generation, result serialization, and the three memory modes before the paid runs. The six completed runs reported no LLM failures. Twenty-four of the 25 questions have evidence inside the supplied 120 turns; the remaining Open-Domain question has no explicit evidence identifier.

8 Conclusion

A-MEM reproduces at the pipeline level. The local ablation gives a positive overall signal for Memory Evolution, with the strongest support on Multi-Hop. The claimed advantage does not transfer consistently to Temporal or Single-Hop under this small DeepSeek subset.

A closer test should expand the question and conversation count, use more seeds, and run the paper-aligned GPT-4o-mini configuration with category-specific retrieval values. Those conditions are required before claiming exact numerical reproduction.